Multi-Task Deep Learning for Breast Lesion Segmentation and Three-Class Classification in Breast Ultrasound Images

Medical Image Analysis
Project Report

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Abstract

Breast cancer is one of the most common cancers worldwide, and ultrasound is a widely available, low-cost imaging tool for breast screening. However, reading ultrasound images is slow and depends on operator experience. This project builds a single deep learning model that performs two tasks at once on breast ultrasound: it segments the lesion at the pixel level and classifies the image into three classes, namely normal, benign, and malignant. Using the public BUSI dataset (780 images), we train four multi-task models that vary the network family (a custom network trained from scratch versus a ResNet34 encoder pretrained on ImageNet) and the use of attention. The best model, the ResNet34-based U-Net, reaches an overall Dice of 0.81, a normal-class recall of 0.95, and a one-versus-rest macro AUC of about 0.95, while keeping a low boundary error (HD95 of 22.7). We further study post-processing and show that class-aware suppression, which clears the predicted mask when the image is classified as normal, improves the overall Dice, whereas largest-component filtering has only a marginal effect. In effect, the network triages a new ultrasound image end to end in a single forward pass: it first decides whether the patient is healthy (normal) or has a lesion, then classifies any lesion as benign or malignant, and at the same time outlines it at the pixel level. The results show that one multi-task model can both localize lesions and recognize healthy images, which is useful as a decision-support tool for radiologists.

1. Introduction

Breast cancer remains a leading cause of cancer death among women, and early detection is the single most important factor for survival. Ultrasound imaging is a key modality in breast examination because it is inexpensive, radiation-free, and especially useful for dense breast tissue where mammography is less reliable. In clinical practice, however, ultrasound images are interpreted manually, which is time-consuming and subject to inter-observer variability. Automated analysis can therefore support radiologists by providing fast, consistent, and reproducible measurements.

This project addresses two clinically meaningful tasks together. The first is lesion segmentation, which produces a pixel-level mask of the suspicious region and allows objective measurement of its size and shape. The second is image-level classification into three categories, normal, benign, and malignant. The three-class formulation is important because it lets the model also recognize a healthy (normal) image, not only distinguish benign from malignant. A practical system must avoid two errors above all: drawing a lesion where there is none, and calling a malignant case healthy.

We combine both tasks in a single multi-task neural network with a shared encoder, a segmentation decoder, and a classification head. To find which design works best, we compare four models that vary the network family and the use of attention. The main contributions of this work are: (i) a multi-task model that performs lesion segmentation and three-class classification simultaneously; (ii) a four-model comparison of two practical network families with and without attention; (iii) a post-processing study that quantifies the effect of largest-component filtering and class-aware suppression; and (iv) a full evaluation with per-class segmentation metrics, boundary distance (HD95), classification metrics with ROC analysis, and paired statistical tests.

2. Background and Related Work

The U-Net is the standard architecture for medical image segmentation. It uses an encoder-decoder structure with skip connections that combine high-level semantic features with low-level spatial detail. The Attention U-Net adds attention gates that re-weight the skip connections so the decoder focuses on relevant regions. A common way to build a strong encoder when the target dataset is small is to start from a network pretrained on the large ImageNet dataset; in this work the pretrained models use a ResNet34 encoder. Multi-task learning trains one model on several related objectives at once, which can act as a regularizer and share useful representations between tasks.

We deliberately use two-dimensional models because the BUSI dataset consists of single two-dimensional frames rather than three-dimensional volumes. Three-dimensional transformer architectures such as UNETR and Swin UNETR are designed for volumetric data and require large training sets, and the nnU-Net framework, although very strong, is heavy to configure and train. These methods are therefore outside the scope of this project and are discussed only as future directions.

3. Dataset

We use the public Breast Ultrasound Images (BUSI) dataset, which contains 780 grayscale ultrasound images collected from female patients, together with expert-annotated segmentation masks. The dataset has three classes: 133 normal images (no lesion), 437 benign images, and 210 malignant images. A small number of lesions are annotated with more than one mask file; these are merged into a single mask using a logical OR. Normal images have an empty mask. The class distribution is clearly imbalanced, which we address with class weights in the classification loss.

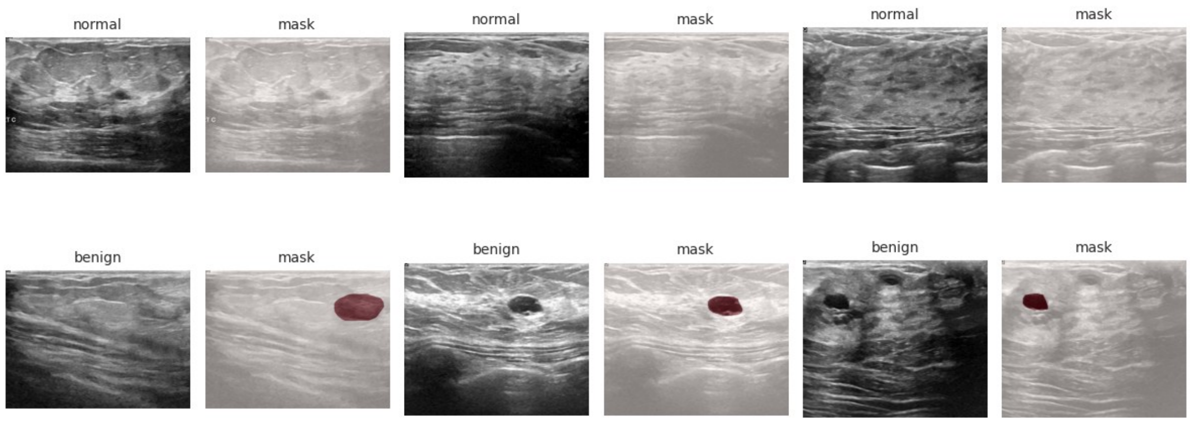

Figure 1. Example BUSI images with their ground-truth masks, shown for the normal and benign classes; malignant examples appear in the qualitative results (Figures 7 and 8).

The images vary in size, so all images are resized to 256 by 256 pixels. An analysis of the lesion area shows that most lesions are small relative to the whole image, which motivates the use of a region-overlap loss (Dice and Tversky) rather than plain pixel accuracy. We split the data into training, validation, and test sets in a stratified 70/15/15 ratio with a fixed random seed, so the class proportions are preserved across the three sets. The test set is held out and used only for the final evaluation.

Subset	Total	Normal	Benign	Malignant
Training	546	≈93	≈306	≈147
Validation	117	≈20	≈65	≈32
Test	117	20	66	31
All	780	133	437	210

Table 1. Stratified data split. The class ratio is preserved in each subset; the test counts are exact.

4. Methods

4.1 Preprocessing and augmentation

All images are resized to 256 by 256 and normalized. Because ultrasound images contain speckle noise, the augmentation pipeline is designed to make the model robust to it. In addition to standard geometric augmentations (horizontal flip and small rotations), we apply CLAHE (contrast-limited adaptive histogram equalization) to enhance local contrast and multiplicative noise to simulate speckle. This treats speckle as a source of variation that the model learns to tolerate, rather than removing it with a separate denoising step.

4.2 Model architectures

We evaluate four concrete multi-task models. Two of them are trained entirely from scratch with our own custom encoder (a plain U-Net and an Attention U-Net), and the other two use a ResNet34 encoder that is pretrained on ImageNet (again a U-Net and an Attention U-Net). The aim of this comparison is not to isolate the effect of pretraining as a single factor; rather, we compare two concrete and commonly used network families, namely a custom from-scratch network and a ResNet34-based network, together with the attention option, in order to find which of these practical choices works best for joint breast-lesion segmentation and classification.

Every model shares the same multi-task structure, illustrated in Figure 2. An encoder turns the input image into feature maps that are shared by two heads. A U-Net decoder uses these features to predict the segmentation mask, while a small classification head attached to the encoder predicts the three-class label. Because the encoder is shared, a single forward pass produces both outputs at once. The classification head is also used at inference time for class-aware suppression: when the image is predicted to be normal, the segmentation mask is cleared (Section 4.5).

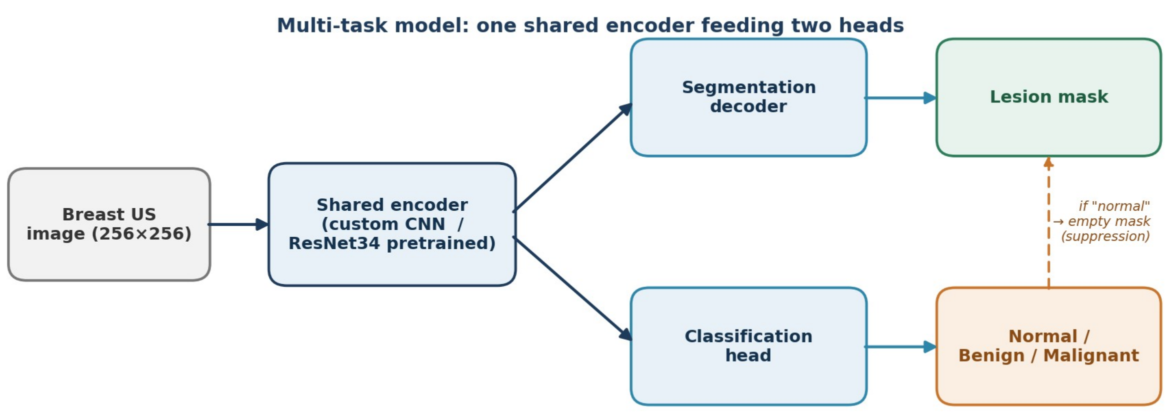

Figure 2. The multi-task architecture. One shared encoder feeds a segmentation decoder (lesion mask) and a classification head (normal / benign / malignant); a normal prediction clears the mask.

4.3 Loss function

The total loss is the sum of a segmentation loss and a classification loss, with the classification term weighted by a factor of 0.4. The segmentation loss combines binary cross-entropy with a Tversky loss; the Tversky parameters (alpha of 0.7 and beta of 0.3) penalize false positives more strongly than false negatives, which reduces scattered false-positive regions. The classification loss is a class-weighted cross-entropy that compensates for the class imbalance described in Section 3.

4.4 Training

All models are trained with the AdamW optimizer and a OneCycle learning-rate schedule (peak learning rate 0.0003) for up to 120 epochs, with a batch size of 32. Early stopping is applied with a patience of 20 epochs based on validation lesion Dice, while final model comparison is reported on the held-out test set using overall Dice as the primary metric. Mixed-precision training is used for speed. A fixed random seed of 42 is used throughout for reproducibility.

4.5 Inference and post-processing

At inference, a probability threshold of 0.5 converts the predicted segmentation map into a binary mask. Two post-processing steps are then optionally applied. Largest-component filtering keeps only the biggest connected region of the predicted mask and removes small spurious blobs. Class-aware suppression uses the classification head: if the model predicts the image to be normal, the segmentation mask is forced to be empty, so no lesion is drawn on a healthy image.

Figure 3 illustrates why connected-component filtering is useful. On the left, the model produces a single clean region that matches the lesion well. On the right, the same model produces one correct main region together with a few small extra regions; these small false-positive blobs lower the score. Largest-component filtering keeps only the main region and discards the extra blobs, which makes the final mask cleaner and easier to use. The quantitative effect of both steps is studied in Section 5.4.

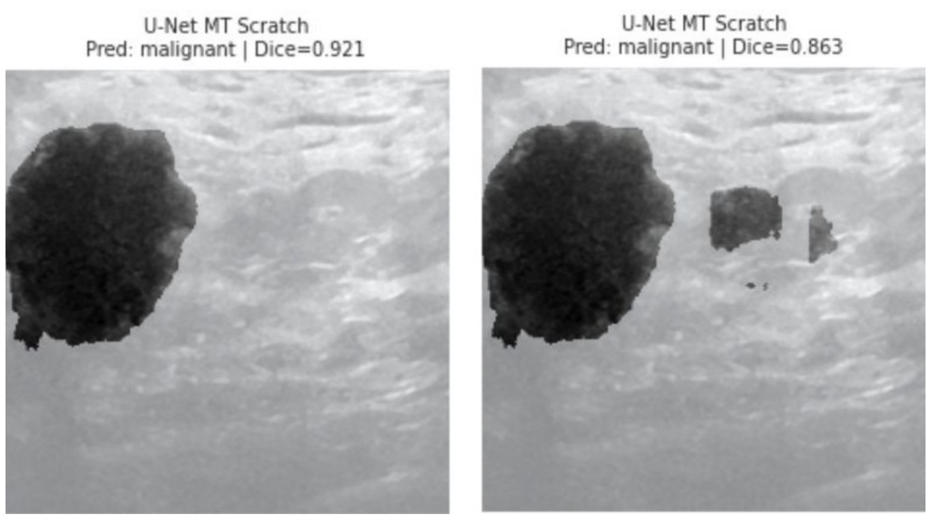

Figure 3. Predicted masks of the from-scratch U-Net. Left: a single clean region (Dice 0.92). Right: a main region with small extra regions (Dice 0.86); largest-component filtering keeps only the main lesion.

4.6 Evaluation metrics

Segmentation quality is measured with the Dice coefficient and the Intersection-over-Union, reported separately for the Overall, Normal, Benign, and Malignant subsets, together with precision and recall. Boundary quality is measured with the 95th-percentile Hausdorff Distance (HD95) using a standard medical-imaging library. Classification is evaluated with accuracy, macro medical-imaging library. HD95 is computed only for samples where a lesion boundary exists; for normal images with empty masks, HD95 is not defined and is excluded from the HD95 average. Classification is evaluated with accuracy, macro F1, per-class one-versus-rest ROC and AUC, and confusion matrices. Differences between models are tested with paired statistical tests on the per-image Dice scores.

5. Results

5.1 Segmentation comparison

Table 2 summarizes the four models on the held-out test set, using overall Dice as the primary metric. Among the four models, the ResNet34-based U-Net (labelled "U-Net pretrained" in the tables) achieves the best overall performance, with the highest overall Dice (0.81), the best normal-subset Dice (0.95), the lowest boundary error (HD95 of 22.7), and the highest macro-F1 (0.852). It improves the overall Dice from 0.779 to 0.809 compared with the custom from-scratch U-Net. The ResNet34-based U-Net also achieves the highest macro-F1, while the two non-attention U-Net models obtain similarly strong macro-AUC values. Adding attention, by contrast, does not improve overall performance in this setting: within each network family the attention variant scores lower on overall Dice, and the ResNet34-based Attention U-Net in particular degrades badly on the normal subset.

Model	Overall Dice	Lesion Dice	Normal Dice	HD95	Acc	Macro-F1	Macro-AUC
U-Net (scratch)	0.779	0.754	0.90	23.6	0.855	0.842	0.948
Attn U-Net (scratch)	0.744	0.732	0.80	29.9	0.778	0.758	0.945
U-Net (pretrained)	0.809	0.780	0.95	22.7	0.855	0.852	0.946
Attn U-Net (pretrained)	0.756	0.788	0.60	23.9	0.744	0.687	0.903

Table 2. Test-set comparison of the four multi-task models with largest-component filtering and class-aware suppression. The ResNet34-based U-Net is selected as the best overall model because it achieves the highest overall Dice, best normal-subset Dice, lowest HD95, and highest macro-F1. Although the pretrained Attention U-Net obtains the highest lesion-only Dice, its normal-class Dice drops substantially.

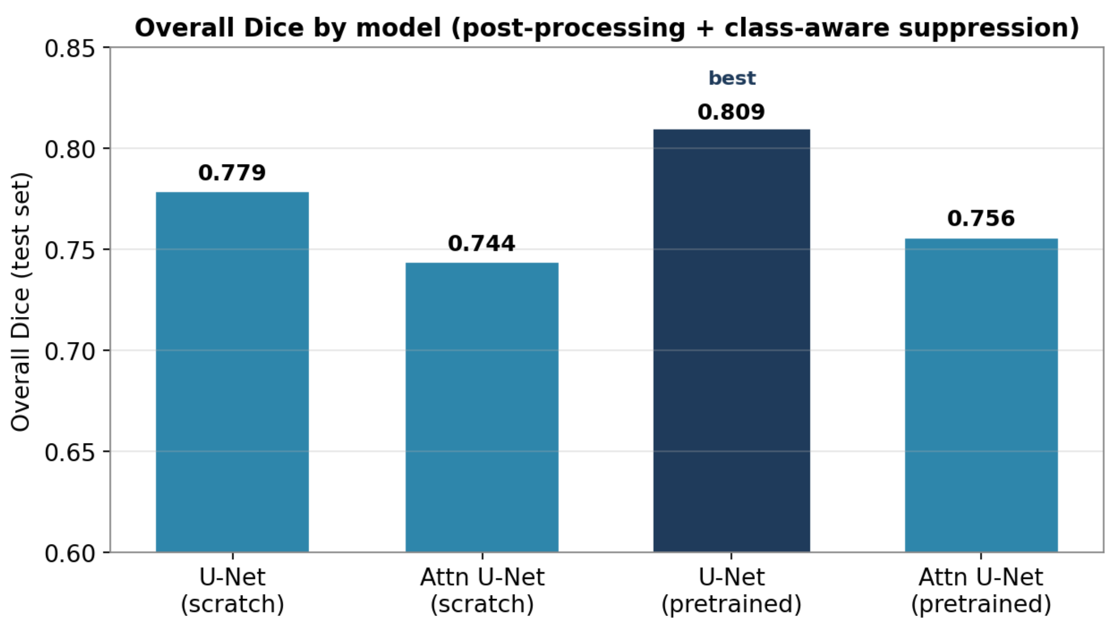

Figure 4. Overall Dice by model. The ResNet34-based U-Net is the best overall model.

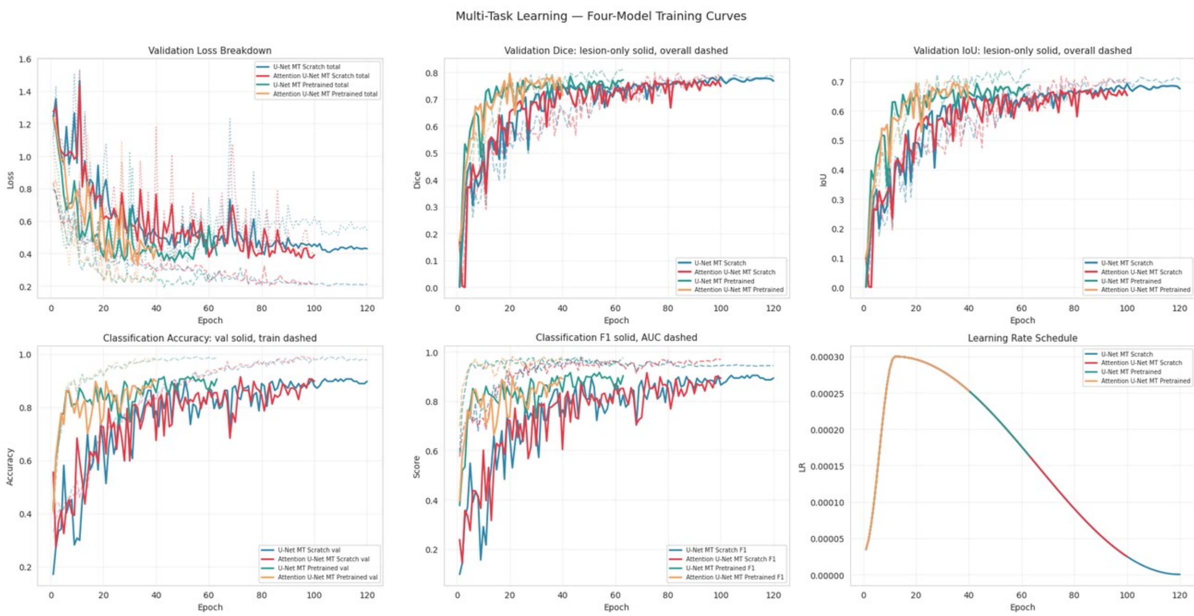

Figure 5. Training and validation curves for the four models, showing stable convergence.

5.2 Per-class segmentation and normal detection

Looking at the per-subset numbers, the ResNet34-based U-Net reaches a Normal Dice of 0.95, meaning that for almost all healthy images the model correctly predicts an empty mask. Its benign and malignant Dice values are 0.80 and 0.73 respectively; malignant lesions are slightly harder, which is expected because they tend to have irregular and blurred boundaries. It is worth noting that, although the ResNet34-based Attention U-Net has a competitive lesion-only Dice (0.788), its Normal Dice collapses to 0.60, which is why its overall Dice is low. This is the main reason we report overall Dice, which balances lesion and normal performance, as the primary metric.

5.3 Three-class classification

Figure 9 shows the per-class ROC curves and confusion matrices for all four models. The ResNet34-based U-Net separates the three classes well, with a one-versus-rest AUC of 0.991 for the normal class, 0.933 for benign, and 0.913 for malignant (macro AUC about 0.95). Its per-class precision, recall, and F1 are given in Table 3. The normal recall is 0.95 (19 of 20 healthy images correctly identified), and the model does not assign the malignant label to any normal image, which is the desired safe behavior. The main remaining error is that a few malignant cases are predicted as benign or normal, which we discuss as a limitation in Section 7.

Class	Precision	Recall	F1-score	AUC (OvR)
Normal	0.86	0.95	0.90	0.991
Benign	0.92	0.85	0.88	0.933
Malignant	0.74	0.81	0.77	0.913
Macro avg	0.84	0.87	0.85	0.946

Table 3. Per-class classification performance of the best model (ResNet34-based U-Net). Overall accuracy is 0.855.

5.4 Effect of post-processing

To understand what the two post-processing steps change, we evaluate the best model under all four on/off combinations. Table 4 reports the overall Dice for every model, and Figure 6 visualizes the effect for the best model. Class-aware suppression is the main contributor: for the ResNet34-

based U-Net it raises the overall Dice from 0.786 to 0.812, because correctly recognized normal images then receive a perfect (empty) prediction. Largest-component filtering, in contrast, has only a marginal and slightly negative effect, since the predicted masks are already fairly clean and the filter can trim a few correct pixels. Using both steps together gives essentially the same result as suppression alone.

Model	None	+ Largest comp.	+ Suppression	Both
U-Net (scratch)	0.771	0.769	0.781	0.779
Attn U-Net (scratch)	0.740	0.740	0.744	0.744
U-Net (pretrained)	0.786	0.783	0.812	0.809
Attn U-Net (pretrained)	0.740	0.748	0.748	0.756

Table 4. Post-processing ablation (overall Dice). Class-aware suppression gives the largest improvement; largest-component filtering is marginal.

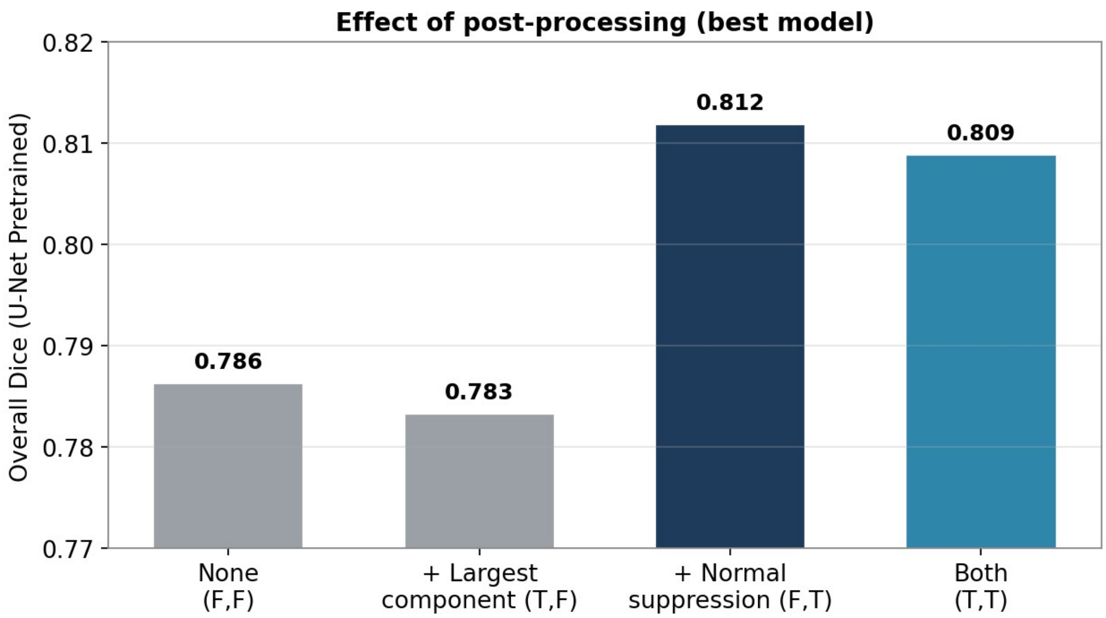

Figure 6. Post-processing ablation for the best model. Suppression is the main contributor.

5.5 Statistical comparison

We compare the models with paired tests on the per-image Dice scores. Attention variants obtain lower overall Dice than their corresponding non-attention models, especially because their normal-class performance is less stable. However, the paired Wilcoxon tests on this single split are not consistently significant for all comparisons. Therefore, the attention result should be interpreted as an observed performance trend rather than a definitive statistical conclusion. The difference between the custom from-scratch U-Net and the ResNet34-based U-Net is positive in favor of the ResNet34 model, but it is also not statistically significant on this single split.

5.6 Qualitative results

Figure 7 shows example predictions of the best model across the three classes. For the normal images the model produces an empty mask and the correct normal label. For the benign and malignant lesions the predicted masks closely match the ground truth, with per-image Dice scores typically between 0.85 and 0.95. The figure also shows a representative error, where one benign lesion is classified as malignant, illustrating the kind of mistake that remains.

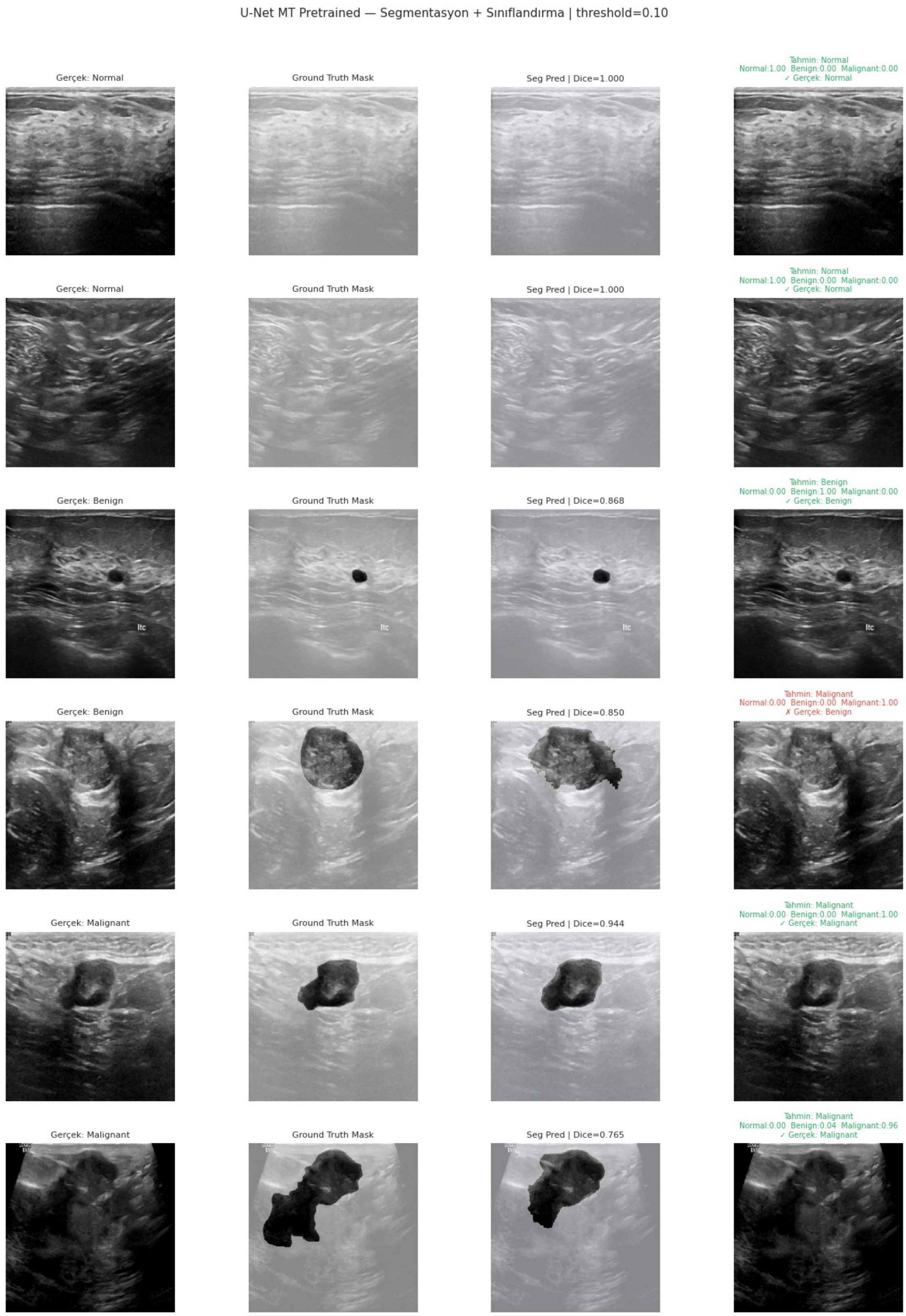

Figure 7. Qualitative predictions of the best model. Columns: input image, ground-truth mask, predicted mask with per-image Dice, and the predicted class (green = correct, red = wrong).

To examine segmentation quality more closely, Figure 8 focuses on three malignant cases, which are the most clinically important and usually the hardest to delineate because of their irregular and blurred boundaries. For each case the predicted mask follows the ground truth very closely, with per-image Dice scores of 0.93 to 0.94. The model captures both the overall extent of the lesion and its uneven, lobulated edges that are typical of malignancy, with only small differences along the contour. This shows that, beyond good average scores, the model produces accurate and usable lesion outlines even on individual difficult cases, which is exactly what a measurement tool needs to be trusted in practice.

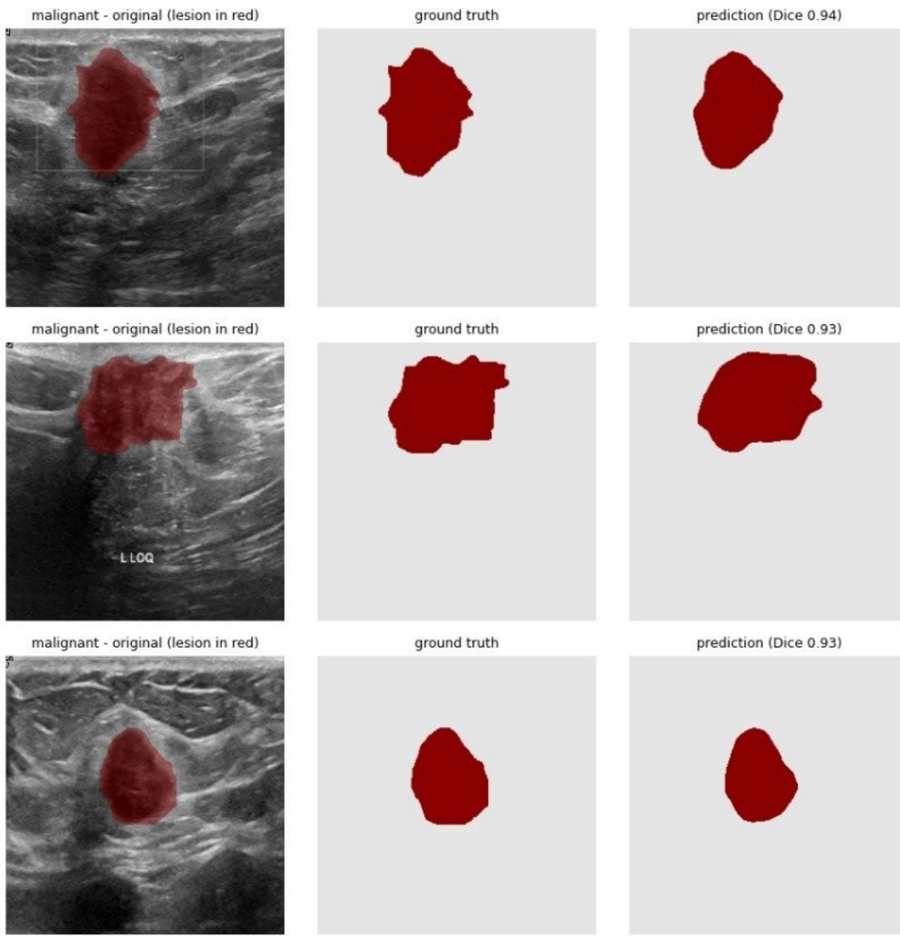

Figure 8. Segmentation of malignant lesions by the best model. Columns: input image with the lesion overlaid in red, ground-truth mask, and predicted mask with the per-image Dice score (0.93 to 0.94).

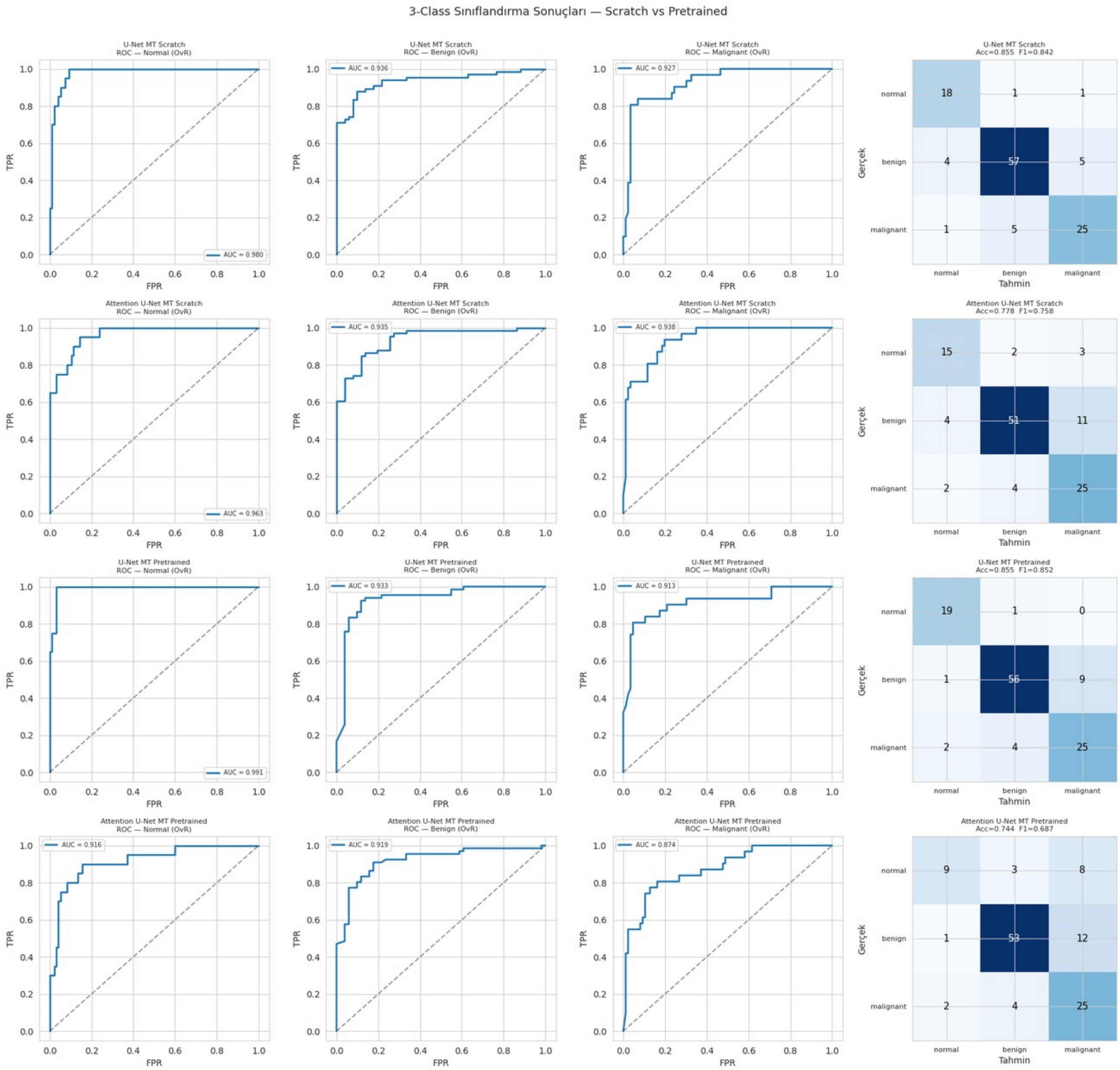

Figure 9. Per-class one-versus-rest ROC curves and confusion matrices for the four models. Rows denote the true labels and columns denote the predicted labels.

6. Discussion

The experiments give a consistent picture. Among the four models, the ResNet34-based U-Net provides the best overall trade-off across segmentation and classification metrics. It improves over the custom from-scratch U-Net in overall Dice, normal-subset Dice, HD95, and macro-F1, although the difference is modest on this single test split. Attention variants did not improve overall performance on this small dataset and reduced normal-class stability, especially in the ResNet34-based Attention U-Net. A likely reason is that the small dataset does not provide enough signal to train the additional attention parameters effectively, so they add noise rather than focus.

It is worth emphasizing what this single multi-task model delivers as a whole. Given a new, unseen ultrasound image, one forward pass answers three questions at once: is the patient healthy or not; if not, is the lesion benign or malignant; and where exactly is it located. A healthy image is recognized as normal and returned with an empty mask, while a lesion image is given a benign or malignant label and is simultaneously outlined at the pixel level. Bringing triage, diagnostic support, and localization together in one model is more efficient and more consistent than running separate detection and classification systems, and it is the central practical strength of the approach.

From a clinical point of view, the most useful property of the system is exactly this combination. The best model identifies 19 of 20 normal images correctly and never labels a normal image as malignant, so it does not raise false cancer alarms on healthy patients. The class-aware suppression step reinforces this behavior by clearing the mask whenever the image is judged normal. At the same time, the system is not perfect: a small number of malignant cases are predicted as benign or normal. Because missing a cancer is the most costly error in this setting, this behavior must be clearly reported and would need to be reduced before any clinical use.

7. Limitations

We list the main limitations of this study so the results are interpreted fairly:

- The dataset is relatively small (780 images) and the split is at the image level rather than the patient level. A patient-level split would be more rigorous and would rule out any chance of images from the same patient appearing in different subsets.
- A few malignant cases are predicted as benign or normal. Missing a malignant lesion is the most serious error in this setting, and reducing it is the priority for any future clinical version.
- The model is trained and tested only on the BUSI dataset. Its behavior on images from other ultrasound devices, hospitals, or patient populations has not been verified, so the results may not transfer directly to a new setting.

8. Conclusion and Future Work

We presented a multi-task deep learning approach that segments breast lesions and classifies ultrasound images as normal, benign, or malignant within a single model. Among four models, the ResNet34-based U-Net is selected as the best overall model, with an overall Dice of 0.81, a normal recall of 0.95, and a macro AUC of about 0.95. In this project, the attention mechanism did not provide the expected improvement: the attention variants failed to improve overall Dice on the small BUSI dataset and reduced normal-class stability. Class-aware suppression, in contrast, remained an effective and simple post-processing step for handling healthy images. The system shows that one model can localize lesions while also recognizing healthy patients, which is a useful basis for a radiologist decision-support tool.

Our future work focuses mainly on moving the model toward real-world use. The most important step is to test and train the model on several different and larger breast-ultrasound datasets, collected from different devices, hospitals, and patient populations, so that it becomes robust across imaging conditions and can be adapted to real clinical settings. Building on this, we plan to keep improving the model itself, for example by calibrating the classifier to further reduce missed malignant cases and by trying stronger backbones, and to integrate it into a practical workflow where a radiologist can quickly review and correct its predictions. More rigorous validation protocols would be applied as part of this real-world adaptation.

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